



Lab 12

Laplace Transform

Name:	Student ID:
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12.1 Objective

The aim of this lab is to investigate the properties of the Laplace transform and apply it to analyze continuous-time Linear Time-Invariant (LTI) systems

12.2 Background

In one of the previous labs, we already have explored the use of the Z-transform for analyzing discrete-time Linear Time-Invariant (LTI) systems. The Z-transform converts discrete-time signals and difference equations into an algebraic form in the z-domain, making it easier to determine system responses, analyze stability, and study system properties such as poles and zeros. Through MATLAB tools, students computed transforms, verified system properties, and examined how pole locations affect the stability of discrete-time systems.

12.3 Introduction

In this lab, the focus shifts to continuous-time systems using the Laplace Transform. The Laplace transform converts time-domain signals and differential equations into the s-domain, allowing continuous-time LTI systems to be analyzed using algebraic techniques. Using MATLAB, students will implement Laplace transforms to derive transfer functions, evaluate system responses, analyze stability through pole locations, and observe how real-world systems such as filters process signals. This lab extends the concepts learned in the previous Z-transform lab by applying similar analysis techniques to continuous-time systems.

Task 1: Computing Laplace Transforms

a) Compute the Laplace transform of the following continuous-time signals using the commands `laplace()`.

1. $x_1[t] = u[t]$
2. $x_2[t] = e^{-2t}u[t]$



3. $x_3[t] = \delta[t]$

b) Use `ilaplace()` to verify the inverse transform.

12.4 The Eigenfunction Property in Continuous Time

A fundamental property of Linear Time-Invariant (LTI) systems is that complex exponentials are eigenfunctions of the system. If the input to a continuous-time LTI system is $x(t) = e^{st}$ the output is $y(t) = H(s)e^{st}$, where $H(s)$ is the transfer function evaluated at that specific complex frequency.

Task 2:

For the continuous time system with a first order differential equation,

$$\frac{dy(t)}{dt} + 0.5y(t) = x(t)$$

The Laplace transform with zero initial condition is,

$$sY(s) + 0.5Y(s) = X(s)$$

where the transfer function provided is,

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+0.5}$$

- a) Define the system using `tf` command.
- b) Apply the input signal $x(t) = e^{j4t}$, to the system
- c) Compute the output using `lsim()` command¹.
- d) Using MATLAB verify that $y(t) = H(j4)x(t) = H(j4)e^{j4t}$.
- e) Plot real part of the simulated output against the theoretical result and also plot the magnitude ratio $\left| \frac{y(t)}{x(t)} \right|$ and verify that they converge as the transient response dies out.

¹ Since this is a continuous time system therefore MATLAB's `filter()` command cannot be used here; as `filter()` function is meant for **discrete-time systems**



Task 3

- For an impulse input signal $x(t)=\delta(t)$ implement the differential equation $\frac{dy(t)}{dt} + 2y(t) = x(t)$ using a time-stepping loop. Assume zero initial conditions.
- Compute $y(t)$ for the first 10-time steps using numerical iteration.
- Now, solve the same system analytically using Laplace Transform and derive $H(s)$.
- For an impulse input compute $Y(s)$ and then use `ilaplace()` to find $y(t)$.
- Create a comparison table showing the first 10 values of:
 - $y(t)$ obtained from the time-stepping loop
 - $y(t)$ obtained from the analytical expression

Task 4

In many practical systems (audio circuits, sensors, communication receivers), signals are corrupted by high-frequency noise. An RC low-pass filter removes this noise.

The transfer function of an RC low-pass filter is

$$H(s) = \frac{1}{RCs + 1}$$

This system allows low frequencies to pass while attenuating high frequencies. Where $R=1\text{Kohm}$ and $C=1\mu\text{F}$

$$x(t) = \sin(2\pi t) + 0.5 \sin(40\pi t)$$

- Generate a signal $y(t)$ consisting of a desired signal $x(t)$ and add noise to the signal and pass the signal through the RC filter using command `lsim()`.
- Plot and compare the original signal $x(t)$ and the filtered output $y(t)$.



Assessment Rubric
Lab 12: Laplace Transform

Name:	Student ID:
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Marks distribution:

	Tasks	LR2 Code	LR5 Results	AR4 Report Submission	AR7 Comments
In-Lab	Task 1	/10	/10	/5	/5
	Task 2	/10	/10		
	Task 3	/10	/10		
	Task 4	/10	/10		
Total =100		/45	/45	/5	/5
CLO Mapped		CLO1	CLO1	CLO1	CLO1

CLO	Total Points	Points Obtained
1	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.